SKYLAR JEAN CALLIS

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EDUCATION

University of Delaware

Working Towards MS in Data Science & PhD in Applied Mathematics

Michigan Technological University (MTU)

BS in Civil Engineering & BS in Applied Mathematics | GPA: 3.97

SKILLS:

Advanced: Python, Git, Google Suite, Public Speaking, Technical Writing

Proficient: Matlab, Wolfram Mathematica, Julia, Linux, LaTeX, Microsoft Office

ACCOLADES

- First place in the 2020 Airport Cooperative Research Program's University Design Competition
- Awarded 2021 Airport Cooperative Research Program's Next Steps Grant
- Awarded 2022 Charles Knobloch Award, 2021 Mathematics Achievement Award, and 2020 Woman of Promise Award by the MTU Mathematical Sciences Department
- Certificates of Merit from the MTU Mathematical Sciences Department in 7 courses

EXPERIENCE

POST-BACHELORS STUDENT IN COMPUTATIONAL PHYSICS (Oct. 2022 - May 2024)

Los Alamos National Laboratory [LANL] (Los Alamos, NM)

- Implemented neural network sensitivity measures to analyze image-to-scalar regression models, with both python's Tensorflow/Keras and Pytorch packages
- Developed and published python module to GitHub, including documentation and unit testing
- Implemented and trained Vision Transformer machine learning models for image-to-scalar regression
- Developed Vision Transformer tutorials and published on GitHub and in the *Towards Data Science* blog
- Presented machine learning research in talks and posters at conferences on Data Science, VVUQ, and High-Performance Computing

DIRECTOR'S SUMMER PROGRAM [DSP] (May - Aug. 2021, seasonal position)

National Security Agency [NSA] (Fort Meade, MD)

- One of 21 participants chosen from 240 applicants nationwide
- Learned about convolutional neural networks and implemented and trained them for novel classification tasks
- Analyzed data from neural network classifiers to improve their performance
- Prepared and delivered briefing to the Director of the National Security Agency on cryptanalytic techniques and results as a result of my research project

SECRETARY & FOUNDING MEMBER OF THE BUILT WORLD ENTERPRISE [BWE] (Jan. 2019 - Dec. 2021)

Michigan Technological University [MTU]'s Enterprise Program (Houghton, MI)

- Developed innovative solutions to address existing problems in the field of civil aviation surface safety, and evaluated these solutions with a cross-disciplinary approach
- Integrated research in human factors, saliency, and situational awareness into a civil engineering foundation
- Gathered interdisciplinary feedback from aviation and human factors industry professionals
- Produced technical writing to describe research practices and results, to persuade for the implementation of innovative solutions, and to apply for external funding